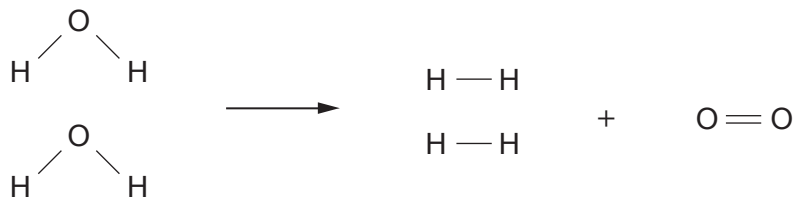


4. The equation shows the bonds which are broken and the bonds which are formed during the electrolysis of water.



- (a) The total energy needed to break the bonds in the reactant is 1856 kJ.

Calculate the energy needed to break **one** O — H bond.

[2]

Energy = kJ

- (b) The total energy released when the bonds in the products are formed is 1370 kJ.

The bond energy for H — H is 436 kJ.

Calculate the energy released when making **one** O = O bond.

[2]

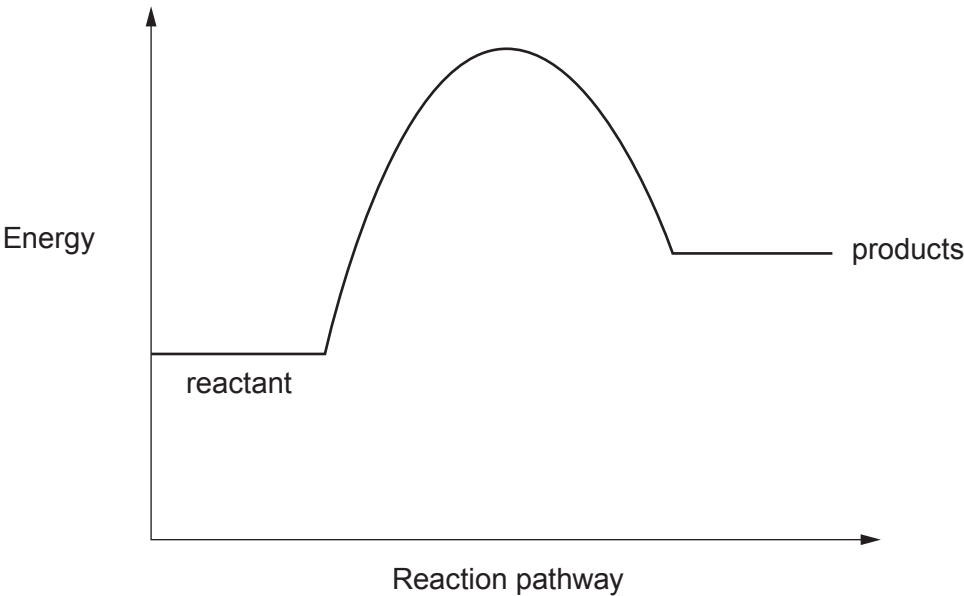
Energy = kJ



(c) The energy profile diagram shows the reaction to be endothermic.

[1]

On the diagram below use the symbol ($\updownarrow$) to show the overall energy change for the reaction.



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